Correct by Construction Concurrent Programs in Idris 2

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About Me

Lecturer at the University of Strathclyde (Glasgow, Scotland)

Interested in:

- Generic Programming and Proving
- Meta Programming and Proof Search
- Type-Directed Partial Evaluation
- Implementations of Type Theory
- Interactive Developer Tooling

Overarching Theme: Correctness by Construction

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Sequential work

One worker mapping the 🪚 transformation on an array.

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🤖

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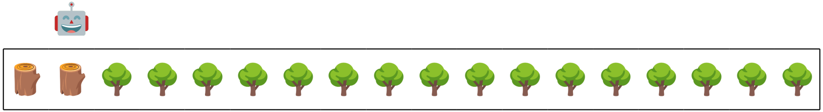

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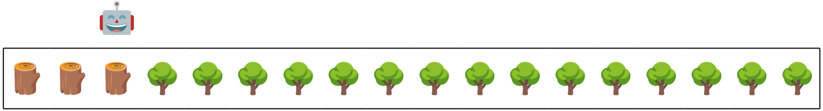

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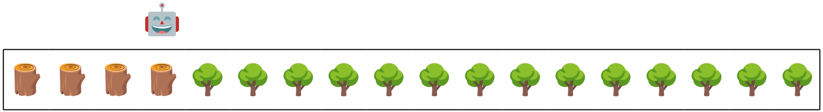

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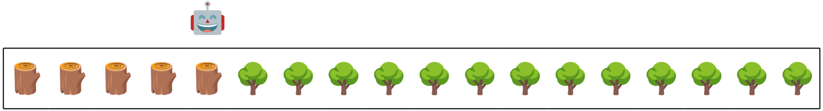

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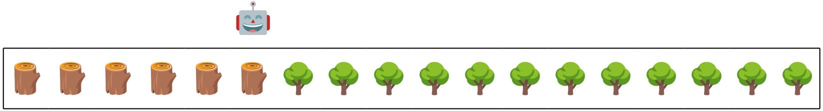

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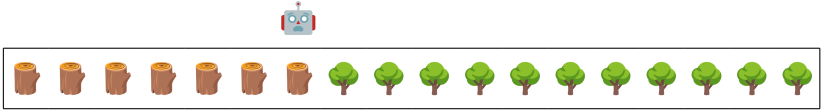

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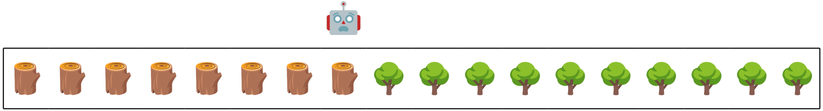

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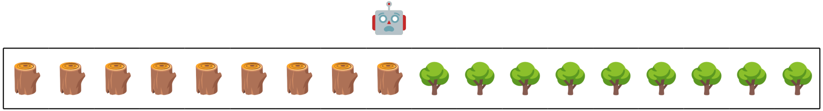

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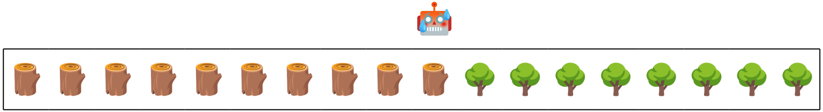

🤖 Can we maybe share the load?

Concurrent work

Three workers mapping the 🪚 transformation on a **shared** array.

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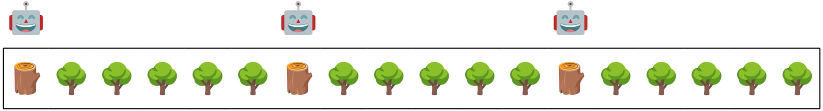

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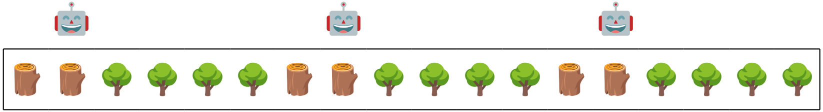

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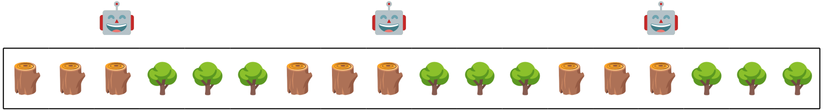

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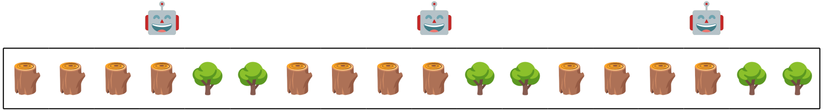

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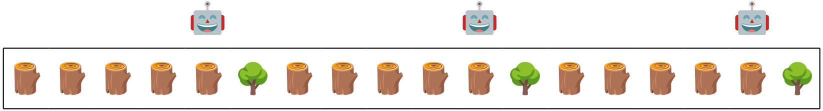

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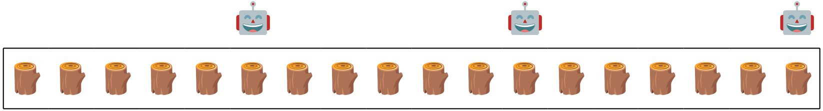

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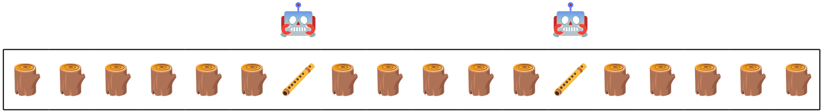

🤖 Faster… but wronger!

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Hoare logic

A logic for imperative programs.
A memory model.
Statements of the form

$$\{P\}c\{Q\}$$

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Assuming that initially P holds

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After executing the command c

$$\{P\}c\{Q\}$$

We can guarantee that Q will hold

Assignment Axiom

$$\{\ell \mapsto \_\} \quad \ell := 0 \quad \{\ell \mapsto 0\}$$

Assignment Axiom

Assuming that ℓ is a valid location

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Ensures that ℓ points to 0

Sequential Execution Axiom

$$\frac{\{P\}c_1\{Q\} \qquad \{Q\}c_2\{R\}}{\{P\}c_1;c_2\{R\}}$$

Sequential Execution Axiom

If c_1 takes us from P to Q

$$\frac{\{P\}\,c_1\,\{Q\} \qquad \{Q\}\,c_2\,\{R\}}{\{P\}\,c_1;c_2\,\{R\}}$$

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Then the composition $c_1;c_2$ takes us from P to R

A proof: swap with no allocation

$$\ell_1 := xor(\ell_1, \ell_2);$$

$$\ell_2 := xor(\ell_1, \ell_2);$$

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A proof: swap with no allocation

$$\{\ell_1 \mapsto a \wedge \ell_2 \mapsto b\}$$

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Combining proofs

$$\frac{\{P\}c_1\{Q\} \qquad \{Q\}c_2\{R\}}{\{P\}c_1;c_2\{R\}}$$

The sequential composition rule is, ironically, anti-compositional: each subprogram needs to talk about the **entire** world no matter what they **actually** use!

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What we *want*: lifting results

$$\frac{\{P\}c\{Q\}}{\{P \wedge R\}c\{Q \wedge R\}}$$

What we *want*: lifting results

$$\frac{\{P\}c\{Q\}}{\{P \land R\}c\{Q \land R\}}$$

If R is also true

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If R is also true

Then R remains true

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If R is also true

Then R remains true

In a sense R is **independent** from P & Q

What we *get*: nonsense!

$$\frac{\{P\}c\{Q\}}{\{P \wedge R\}c\{Q \wedge R\}}$$

$$\{\ \ell \mapsto 1\ \}\ \ \ell := 0\ \ \{\ \ell \mapsto 0\ \}$$

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$$\left\{ \begin{array}{l} \quad\ \ell \mapsto 1 \\ \wedge \end{array} \right\} \ \ell := 0 \ \left\{ \begin{array}{l} \quad\ \ell \mapsto 0 \\ \wedge \end{array} \right\}$$

What we *get*: nonsense!

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We have: $P = \ell \mapsto 1$ and $Q = \ell \mapsto 0$
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🤖 0 is equal to 1?!

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🤖 0 is equal to 1?!

Nothing actually enforces that R is **independent** from P & Q!

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What we *build*: a separating conjunction

- Make predicates support-aware
- Disallow claims over overlapping memory regions

	Predicate	Support

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What we *build*: a separating conjunction

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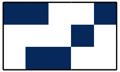
	Predicate	Support
Purely logical	$m + n = 3$	
Points to	$\ell \mapsto v$	
Conjunction	$P * Q$	?

What we *build*: a separating conjunction

- Non-overlapping:

 *  =

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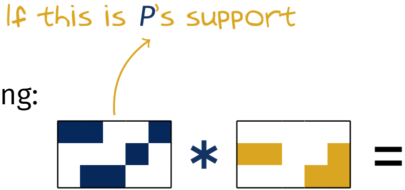
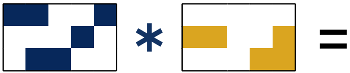


What we *build*: a separating conjunction

If this is P's support

and this Q's support

- Non-overlapping:



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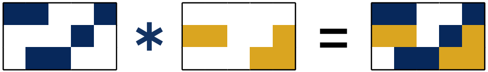

Then $P * Q$ is defined as $P \wedge Q$ and has this support

What we *build*: a separating conjunction

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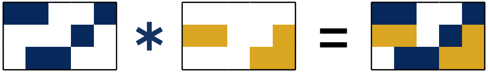
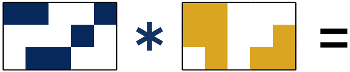


- Overlapping



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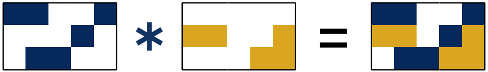

If this is P's support

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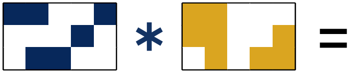

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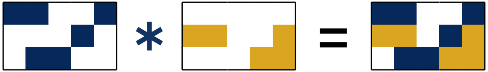

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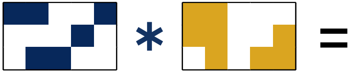

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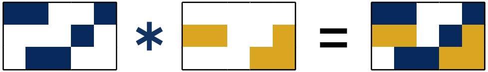

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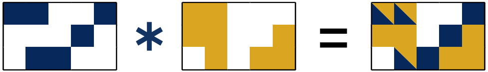

Then this is not a valid support.

$P * Q$ collapses to the absurd predicate $\perp$

What we *obtain*: the frame rule

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

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If R is true and non-overlapping

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$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

If R is true and non-overlapping

Then R remains true and non-overlapping

Revisiting our footgun

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

$$\{\ \ell \mapsto 1\ \}\ \ell := 0\ \{\ \ell \mapsto 0\ \}$$

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We have: $P = \ell \mapsto 1$ and $Q = \ell \mapsto 0$

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Revisiting our footgun

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

We have: $P = \ell \mapsto 1$ and $Q = \ell \mapsto 0$
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Revisiting our footgun

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

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Overlapping! $P * R$ and $Q * R$ both equal $\perp$

Revisiting our footgun

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

We have: $P = \ell \mapsto 1$ and $Q = \ell \mapsto 0$
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$$\{\perp\}\ \ell := 0\ \{\perp\}$$

Revisiting our footgun

$$\frac{\{P\}c\{Q\}}{\{P * R\}c\{Q * R\}}$$

We have: $P = \ell \mapsto 1$ and $Q = \ell \mapsto 0$
We pick: $R = \ell \mapsto 1$

$$\{\bot\} \ \ \ell := 0 \ \ \{\bot\}$$

🤖 Garbage in; garbage out

From Points-to to Ownership

$$\ell \mapsto v$$

Meaning:

- used to be "ℓ points to v"
- now is "I **own** ℓ and it points to v"

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Ownership:

- is globally unique
- is transferrable
- allows destructive updates

From Points-to to Ownership

$$\ell \mapsto v$$

Meaning:

- used to be "ℓ points to v"
- now is "I **own** ℓ and it points to v"

Ownership:

Somewhat paradoxically, this allows **local** reasoning

- is globally unique
- is transferrable
- allows destructive updates

From Points-to to Ownership

$$\ell \mapsto v$$

Meaning:

- used to be "ℓ points to v"
- now is "I **own** ℓ and it points to v"

Ownership:

- is globally unique
- is transferrable
- allows destructive updates

All of this is implicitly enforced by the rules of the logic

Table of Contents

Old School Verification: Write, Test, Fix loop

```
10 WRITE CODE
20 DO FORMALISATION
30 IF (CONTAINS BUG) THEN
40    GOTO 10
50 END IF
```

Correct by Construction: Specify, Implement Correctly, Keep

Sometimes known as goal-driven development

1. Write a specification
2. In a dialogue with the compiler interactively refine it
   * Each step produces part of the program
   * Some step introduce some further goals too
3. Keep refining until all goals are trivials

In This Talk: Idris 2

- Functional (lambdas, pure functions, inductive types)

```
swap : (a, b) -> (b, a)
swap = \ (x, y) => (y, x)
```

In This Talk: Idris 2

- Functional (lambdas, pure functions, inductive types)
- First class types (i.e. types are standard values)

```
FileLoc : Type
FileLoc = (String, Nat, Nat)
```

In This Talk: Idris 2

- Functional (lambdas, pure functions, inductive types)
- First class types (i.e. types are standard values)
- Resource-aware (separation of specification vs. runtime)

```
id : {0 a : Type} -> a -> a
id x = x
```

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```
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Quantity 0: erased during compilation

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- Strict (with explicit Laziness annotations)

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- Self-hosted (reasonably fast!)

In This Talk: Core Idea

Define a Domain Specific Language internalising Separation logic ideas

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- Linearity (ab)used to ensure global uniqueness

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Define a Domain Specific Language internalising Separation logic ideas

- Linearity (ab)used to ensure global uniqueness
- Ownership proofs instead of raw pointers
- Erasure to get rid of specification data (values showing up in Ps, Qs, Rs)

Ownership Type

$$region[start, end] \mapsto vs$$

```
data Owned :
  (region : Region) -> (start, end : Nat) ->
  (vs : List Bits8) -> Type where
```

Read

$$\left\{ \qquad\qquad\qquad\qquad \right\}$$

$$v = \texttt{getBits8}(idx);$$

$$\left\{ \qquad\qquad\qquad\qquad \right\}$$

Read

$$\left\{ \quad region[start, end] \mapsto vs \quad \right\}$$

$$v = \texttt{getBits8}(idx);$$

$$\left\{ \qquad \right\}$$

Read

$$\left\{ \begin{array}{rl} & region[start, end] \mapsto vs \\ * & 0 \leq idx < |vs| \end{array} \right\}$$

$$v = \texttt{getBits8}(idx);$$

$$\left\{ \qquad \right\}$$

Read

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$$v = \texttt{getBits8}(idx);$$

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Read

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$$v = \texttt{getBits8}(idx);$$

$$\left\{ \begin{array}{ll} & region[start, end] \mapsto vs \\ * & v = vs[idx] \end{array} \right\}$$

```
getBits8 :
  LinearIO io =>
  {start, end : Nat} ->
  (1 _ : Owned region start end vs) ->
  (idx : Nat) -> (0 _ : InBounds idx vs) ->
  L1 io (WithVal (Owned region start end vs)
                 (Singleton (index idx vs)))
```

Write

$$\left\{ \qquad\qquad\qquad\qquad \right\}$$

$\texttt{setBits8}(idx, val);$

$$\left\{ \qquad\qquad\qquad\qquad \right\}$$

Write

$$\left\{ \quad region[start, end] \mapsto vs \quad \right\}$$

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Write

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Write

$$\left\{ \begin{array}{rl} & region[start, end] \mapsto vs \\ * & 0 \leq idx < |vs| \end{array} \right\}$$

$$\texttt{setBits8}(idx, val);$$

$$\left\{ \quad region[start, end] \mapsto vs[idx := val] \quad \right\}$$

Write

$$\left\{ \begin{array}{ll} & region[start, end] \mapsto vs \\ * & 0 \leq idx < |vs| \end{array} \right\}$$

$$\texttt{setBits8}(idx, val);$$

$$\left\{ \quad region[start, end] \mapsto vs[idx := val] \quad \right\}$$

```
setBits8 :
  LinearIO io =>
  {start : Nat} ->
  (1 _ : Owned region start end vs) ->
  (idx : Nat) -> (0 _ : InBounds idx vs) ->
  (val : Bits8) ->
  L1 io (Owned region start end (replaceAt idx val vs))
```

Split

$$\left\{ \qquad\qquad\qquad\qquad\qquad\qquad \right\}$$

$\texttt{splitAt}(m);$

$$\left\{ \qquad\qquad\qquad\qquad\qquad\qquad \right\}$$

Split

$$\left\{ \quad region[start, end] \mapsto vs +\!\!+\, ws \quad \right\}$$

$\texttt{splitAt}(m);$

$$\left\{ \qquad \right\}$$

Split

$$\left\{\begin{array}{rl} & region[start, end] \mapsto vs +\!\!+\, ws \\ * & |vs| = m \end{array}\right\}$$

$$\texttt{splitAt}(m);$$

$$\left\{ \quad \right\}$$

Split

$$\left\{ \begin{array}{ll} & \mathit{region}[\mathit{start}, \mathit{end}] \mapsto \mathit{vs} \mathbin{+\!\!+} \mathit{ws} \\ * & |\mathit{vs}| = m \end{array} \right\}$$

$$\texttt{splitAt}(m);$$

$$\left\{ \begin{array}{ll} & \mathit{region}[\mathit{start}, \mathit{start} + m] \mapsto \mathit{vs} \\ * & \mathit{region}[\mathit{start} + m, \mathit{end}] \mapsto \mathit{ws} \end{array} \right\}$$

Split

$$\left\{ \begin{array}{ll} & region[start, end] \mapsto vs ++ ws \\ * & |vs| = m \end{array} \right\}$$

$$\texttt{splitAt}(m);$$

$$\left\{ \begin{array}{ll} & region[start, start + m] \mapsto vs \\ * & region[start + m, end] \mapsto ws \end{array} \right\}$$

```
splitAt :
  {0 vs, ws : List Bits8} ->
  {m : Nat} -> (0 _ : HasLength m vs) ->
  Owned region start end (vs ++ ws) -@
  LPair (Owned region start (start + m) vs)
        (Owned region (start + m) end ws)
```

Combine

$$\left\{ \vphantom{\begin{array}{c}a\\b\end{array}} \qquad\qquad\qquad\qquad \right\}$$

$\texttt{combine}();$

$$\left\{ \qquad\qquad\qquad\qquad \right\}$$

Combine

$$\left\{ \begin{array}{rr} & region[start, middle] \mapsto vs \\ * & region[middle, end] \mapsto ws \end{array} \right\}$$

$$\texttt{combine}();$$

$$\left\{ \qquad \right\}$$

Combine

$$\left\{ \begin{array}{rl} & region[start, middle] \mapsto vs \\ * & region[middle, end] \mapsto ws \end{array} \right\}$$

$$\texttt{combine}();$$

$$\left\{ \quad region[start, end] \mapsto vs \mathbin{+\!\!+} ws \quad \right\}$$

Combine

$$\left\{ \begin{array}{rl} & \mathit{region}[\mathit{start}, \mathit{middle}] \mapsto \mathit{vs} \\ * & \mathit{region}[\mathit{middle}, \mathit{end}] \mapsto \mathit{ws} \end{array} \right\}$$

$$\texttt{combine}();$$

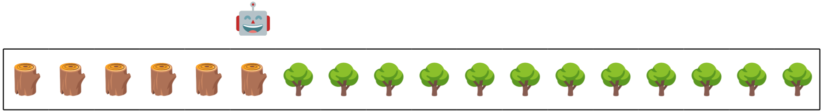
$$\left\{ \mathit{region}[\mathit{start}, \mathit{end}] \mapsto \mathit{vs} \mathbin{+\!\!+} \mathit{ws} \right\}$$

```
(++) :
  Owned region start middle vs -@
  Owned region middle end ws -@
  Owned region start end (vs ++ ws)
```

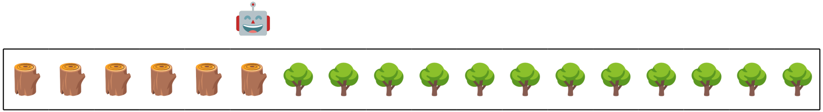
Map Type

```
Map : (Type -> Type) -> Type
Map io =
  forall region. {start, end : Nat} ->
  {0 trees : List Bits8} ->
  (saw : Bits8 -> Bits8) ->
  (1 _ : Owned region start end          trees) ->
  L1 io (Owned region start end (map saw trees))
```

Sequential Map - Loop Type

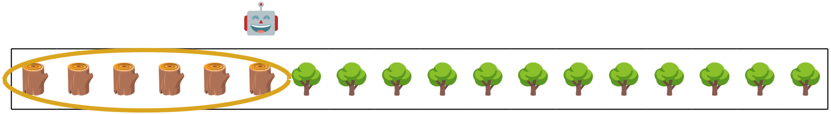


Sequential Map - Loop Type



```
(1 _ : Owned region start end ((map saw treesL) <>> treesR)) ->
L1 io (Owned region start end (map saw (treesL <>> treesR)))
```

Sequential Map - Loop Type



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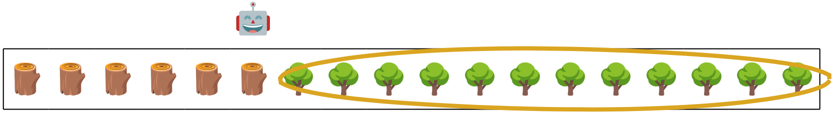

```
(1 _ : Owned region start end ((map saw treesL) <>> treesR)) ->
L1 io (Owned region start end (map saw (treesL <>> treesR)))
```

Parallel Map

Parallel Map

```
halve :
  {start, end : Nat} ->
  (1 _ : Owned region start end trees) ->
  Res Nat (\ m =>
  LPair (Owned region start         (start + m)  (take m trees))
        (Owned region (start + m) end            (drop m trees)))
```

Parallel Map

```
halve :
  {start, end : Nat} ->
  (1 _ : Owned region start end trees) ->
  Res Nat (\ m =>
  LPair (Owned region start        (start + m)  (take m trees))
        (Owned region (start + m) end           (drop m trees)))
```

```
par1 : L1 IO a -@ L1 IO b -@ L1 IO (LPair a b)
```

Parallel Map

```
halve :
  {start, end : Nat} ->
  (1 _ : Owned region start end trees) ->
  Res Nat (\ m =>
  LPair (Owned region start          (start + m)  (take m trees))
        (Owned region (start + m) end             (drop m trees)))
```

```
par1 : L1 IO a -@ L1 IO b -@ L1 IO (LPair a b)
```

```
parMapRec : Map IO -> Map IO
parMapRec subMap saw buf
   = do let (m # lbuf # rbuf) = halve buf
        (lbuf # rbuf) <- par1 (subMap saw lbuf) (subMap saw rbuf)
        let 1 buf = lbuf ++ rbuf
        pure1 (reindex (mapTakeDrop saw m trees) buf)
```

Parallel Reduce

Apply the same principles to get a parallel reduce
Relying on monoid laws to prove correctness

What's next?

Separation logic has a lot more to offer!

- Partial ownership (shared reads, owned writes)
- Locks (non-deterministic access to shared resources)
- Ghost states (stateful specification data)

Use these building blocks!

- Richly typed parallel skeletons
- Reintroduce layers of abstractions (e.g. inductive types)
- Seamless programming over serialised data
- Concurrent programs

Happy to Chat! See You in Glasgow?

https://gallais.github.io

https://mamot.fr/@gallais

TYPES 2025 — 9–13 June

Glasgow, Scotland

https://msp.cis.strath.ac.uk/types2025/